A deterministic emulator of the Veros ACC last-year state

31 October 2026

Summary

- Goal: predict the temperature of the last year of 100-year Veros ACC runs directly from the two constants (c_k, c_ϵ) of the TKE vertical-mixing closure, without running the model. The target is the state at year 100 of this protocol, not an equilibrium (Section 1).
- Method: a small deterministic network maps $z = (\log c_k, \log c_\epsilon)$, standardised, to the leading PCA coefficients of the training states; it is an ensemble of 5 initialisations. Hyper-parameters come from repeated cross-validation with the one-standard-error rule. A fit takes seconds on a CPU.
- The two constants act mainly through $s \approx \log c_k - 1.08 \log c_\epsilon$, close to their ratio, as the closure's scaling invariance implies (Section 1). The emulators that use s are the strong ones.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks, leave-one-out): RMSE 0.017–0.038 K.
  - A Gaussian process on s, the best GP tested, gives 0.019–0.045 K. The network's mean is lower on all five evaluations, by 0.0025–0.0075 K (13–17 %).
  - This margin is consistent but not statistically established. Per-run tests give Holm $p \leq 0.044$ on four of the five, but runs on one c_k/c_ϵ diagonal are near-replicates, so those tests are optimistic. With diagonals as units no evaluation is significant†.
  - Not compared here: linear interpolation of the fields along s and a GP on $(s, s_\perp)$. Linear interpolation along s appears to match the network in interpolation†.
  - The nearest run in z gives 0.038–0.118 K; its value depends on how equidistant neighbours are ordered.
- Extrapolation (top row, outer 51, outer 75): 0.049 / 0.067 / 0.114 K.
  - The GP on s gives 0.079 / 0.074 / 0.125 K. Per-run tests favour the network on outer 51 and outer 75 (Holm $p = 0.013$ and 0.004; on outer 75 only with the one-standard-error choice). With diagonals as units neither is significant†.
  - The cause differs by split†. On outer 75 nearly all of the margin comes from runs beyond the warm end of the training range in s, where the network continues the steep warm-end slope and the GP flattens. On outer 51 and the top row it comes mostly from the second direction $s_\perp$.
  - The error is largest along the warm edge (large c_k, small c_ϵ). At the warm corner it is 0.16 K one row from the training set and 1.06 K three cells away.
  - Extrapolated fields toward the warm corner contain temperature inversions of up to several kelvin, against at most 0.17 K in the truth, for every method†. With this equation of state they are density inversions. They come from the 20-mode decoder, so such fields are not usable model states as they stand.
- The margin over the GP on s also depends on the selection rule. With the minimum CV error instead, the network ties on the 7×7 block and on outer 75. The one-standard-error rule was adopted after a first review had seen test errors, so its margins are not a blind test. On the eight main evaluations the network is below the GP on z, the RBF and the nearest run under both rules.
- Robustness: four initialisation sets move a split's mean by 0.0004–0.0010 K, and by 0.0028 K on the top row. Among the configurations that cross-validation cannot tell apart, the test error spans up to 0.03 K. With one BLAS thread a refit reproduces every number bit for bit.
- With four training runs cross-validation cannot choose the hyper-parameters. Depending on them, the network gets 0.10–0.19 K, against 0.120 K for the GP on s (Section 6).

†Found in an independent review with scripts outside this repository; not yet reproduced in `det/` (Section 7).

1 Setup

- Model: mini_veros, a JAX port of Veros [8, 9] by E. Meunier, with its ACC setup (`acc/full`), which uses the settings of Veros's `setups/acc/acc.py`. Spherical grid of 2° cells, 30 in longitude and 42 in latitude (about 42°S to 42°N), 15 levels down to 2080 m, flat bottom.
- Geometry: a meridional wall, 4° wide, from 20°S to the northern boundary. South of 20°S the domain is a re-entrant channel; north of it, a closed basin.
- Forcing, steady: a zonal wind stress of up to $0.1\,\mathrm{N\,m^{-2}}$ over the channel and a second band, of up to $0.2\,\mathrm{N\,m^{-2}}$, north of 10°N; surface temperature restored with a 30-day time scale to a profile that is 15 °C between 20°S and 20°N and falls to 0 °C at both ends.
- Initial state: T decreasing linearly with depth from 15 °C; salinity 35 everywhere. The equation of state is Veros type 3, nonlinear, with density decreasing with T over the whole range of the data.
- Closures, as in the setup: TKE vertical mixing [6] with the mixing length limited by $\sqrt{2e}/N$ [1], minimum diffusivities $\kappa_H = 2\times10^{-5}\,\mathrm{m^2s^{-1}}$ and $\kappa_M = 2\times10^{-4}\,\mathrm{m^2s^{-1}}$; neutral diffusion and GM thickness diffusion [7] with K_{gm} from the EKE closure [5], held at its setup values $(0.4, 0.5)$; no IDEMIX.
- Varied: the TKE constants. With mixing length l and turbulent kinetic energy e,

$$K_M = c_k\, l\sqrt{e}, \qquad \epsilon = c_\epsilon\, e^{3/2}/l, \qquad l = \sqrt{2e}/N. \tag{1}$$

  The setup default is $(0.1, 0.7)$, the geometric centre of the design (Figure 1).
- Why one combination: with $l = \sqrt{2e}/N$, both $K_M = \sqrt{2}\,c_k e/N$ and $\epsilon = c_\epsilon N e/\sqrt{2}$ are unchanged under $(c_k, c_\epsilon, e) \to (\lambda c_k, \lambda c_\epsilon, e/\lambda)$. A steady state then depends on c_k/c_ϵ only, apart from what breaks the invariance in the closure's equations: the vertical diffusion of e (proportional to $K_M e$), and the limits on l (the distance to the boundaries and a minimum), where l no longer follows $\sqrt{e}$. The runs are also not steady. The sources of e and the bounds on K_M and κ_H (minimum, maximum, depth profile) keep it, since they act on invariant quantities; the minimum diffusivities may set the flat response at small c_k/c_ϵ (not tested). The ratio is proportional to a critical Richardson number, and it dominates the global mixed-layer depth in Veros [13].
- Check on the raw runs (last output, median over the water cells of levels 3–11). Along the diagonal $c_k/c_\epsilon = 1/7$, from $c_k = 0.0125$ to 0.504, e falls about 45-fold while $c_k e$ stays at 1.7–$1.9\times10^{-7}\,\mathrm{m^2s^{-2}}$, and the EKE stays at $7.5\times10^{-4}\,\mathrm{m^2s^{-2}}$. So the varied constants are the TKE ones. The sweep's own records (MiniVeros-Diffusion repository) also describe c_k and c_ϵ as the TKE closure's constants.

2 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$ (ratio $2^{2/3}$); 100 years, a snapshot every 30 days; interior grid $30\times42\times15$ (x, y, z) with the 930 cells of the wall.
- Target: $y_r \in \mathbb{R}^{17970}$, the mean temperature of the last 12 snapshots (360 days) of run r on the water cells. $S = 35$ psu everywhere, so it is dropped.
- The runs still drift, so the averaging window matters. Over the last 20 years, the RMS change from the first to the last year is 0.16 K, and the last-year mean differs from the 20-year mean by 0.08 K. The target is therefore defined by this protocol (100 years from the setup's initial state), and every method is scored on the same definition.
- The two parameters act mainly through one combination, $s = \log c_k - 1.08\,\log c_\epsilon$ (Figure 1). The exponent is empirical, from a linear fit of the first PCA coefficient; it is close to the value 1 of the closure's invariance (Section 1).
- Below $s \approx -2$ (48 runs) the state is nearly flat in s: the domain-mean T stays within 0.09 K. Above, it rises by 1.1 K. Split means therefore mix a flat and a warm regime.

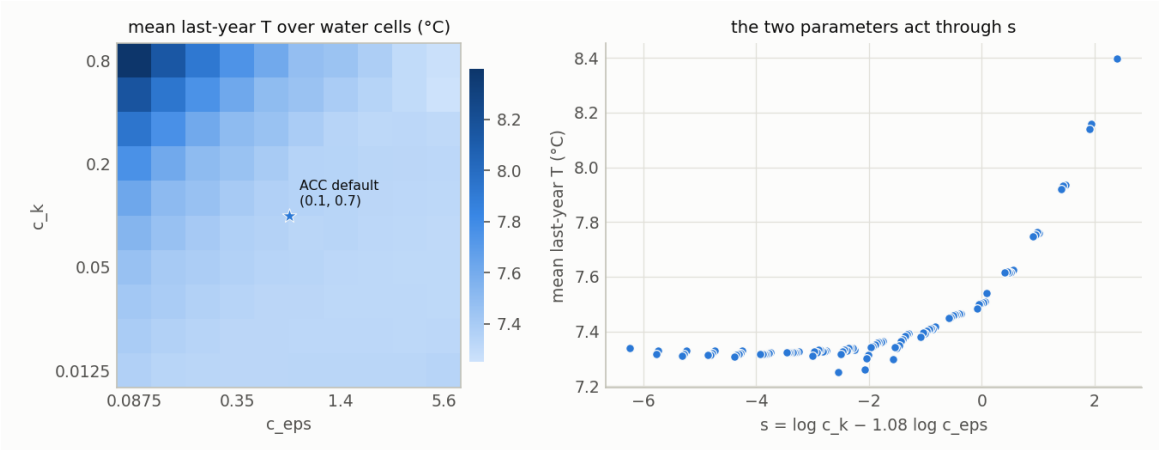


Figure 1: Mean last-year T (°C) over the water cells: over the parameter grid (left; star: the setup default) and against s (right), the direction along which the first PCA coefficient of the states varies linearly.

3 Method

- Input z: $(\log c_k, \log c_\epsilon)$ standardised over the design.
- Output: a field in the span of the training states,

$$\hat{y}(z) = \mu + \sum_{k=1}^{K} a_k(z)\, v_k, \qquad K = \min(20,\, n-1), \tag{2}$$

  where μ and v_k are the mean and the leading principal directions of the n training states. So $K = 20$ on every split except the four-run one ($K = 3$).
- This is the standard construction of a PCA-basis emulator [11], with a network in place of the usual GPs.
- Network $a(z)$: a multilayer perceptron $2 \to 64 \to 64 \to K$ (scikit-learn [14], L-BFGS). Its targets are $a_k(z_r) = v_k^\top (y_r - \mu)$ divided by $\mathrm{std}(a_1)$, so with orthonormal v_k the data term is the physical MSE of the projected field up to a constant. In scikit-learn's normalisation:

$$\mathcal{L} = \frac{1}{2nK} \sum_{r=1}^{n} \frac{\|a(z_r) - V^\top (y_r - \mu)\|^2}{\mathrm{std}(a_1)^2} + \frac{\alpha}{2n} \|W\|^2. \tag{3}$$

- Ensemble: the prediction is the mean of 5 networks that differ only in their initialisation. The whole evaluation is repeated with 4 such sets, and reported values average them.
- Hyper-parameters: $(\text{activation}, \alpha) \in \{\text{ReLU}, \tanh\} \times \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, chosen on the training runs of each split.
  - The search is 5-fold cross-validation repeated 3 times. A configuration's CV RMSE is its RMSE per training run, averaged over the repeats and then over the runs.
  - The one-standard-error rule [10] then picks the most regularised configuration whose CV RMSE is within one standard error of the best. That standard error is over runs: the standard deviation of the best configuration's per-run CV RMSE, over $\sqrt{n}$. It is not the usual standard error over folds, and the choice of standard error can change the selection [2].
  - The choice is ReLU with $\alpha = 10^{-2}$ on every split except outer 75 ($\alpha = 10^{-1}$) and four runs ($\alpha = 1$).
  - The minimum of the same scores is ReLU with $\alpha = 10^{-3}$ or 10^{-4}, except on outer 75 (10^{-2}) and four runs (1).
  - The rule is not blind to the test sets. It replaced a single 5-fold cross-validation that took the minimum, after an independent review had shown that choice to be decided by fold noise, and that review had already computed test errors of other configurations. Table 2 therefore also gives the network chosen by the minimum of the same cross-validation scores.

  - For leave-one-out the choice is made once, on all 100 runs. Each held-out run therefore takes part in it: a mild selection leak, with flat scores.
- Fitting on run means rather than on snapshots changes nothing: $\sum_t \|f(z_r) - x_{r,t}\|^2 = n_t \|f(z_r) - \bar{x}_r\|^2 + \text{const}$.
- Baselines, none tuned by cross-validation (unlike the network):
  - the training mean;
  - the nearest training run in z. The grid is equally spaced in both standardised inputs, so an interior held-out run often has two to four equidistant training neighbours; the baseline takes the first in storage order, and its error depends on that order;
  - thin-plate RBF interpolation of every cell (exact, no smoothing);
  - a GP per PCA coefficient on z (Matérn 5/2, one length scale per input, white noise up to the full variance; hyper-parameters by maximum marginal likelihood [15]);
  - the same GP on $s = w^\top z$, with w from a linear fit of a_1 on z within the training runs: a one-dimensional active subspace [3].

4 Evaluation

- Eight hold-out splits of the grid, plus leave-one-out over the 100 runs:
  - interpolation: 10 scattered interior runs; the rows $c_k = 0.126$, 0.2, 0.3175; the centre 5×5 and 7×7 blocks;
  - extrapolation: the top row ($c_k = 0.8$); the 51 or 75 outer runs, training on the centre;
  - four training runs, one step in from each corner (Section 6).
- Metric: RMSE over the water cells (each cell weighted equally, not by volume) against the last-year mean, per held-out run, then the mean and median over the runs.
- Paired comparison with the GP on s, on the per-run difference (network RMSE averaged over the 4 initialisation sets, minus the GP's): its mean, a 95 % bootstrap interval, and a Wilcoxon signed-rank p-value (exact for up to 50 runs, normal approximation above). The p-values are also given with a Holm correction [12] over the eight main evaluations (the four-run split excluded).
  - These tests treat the held-out runs as independent. They are not: runs on one c_k/c_ϵ diagonal have nearly the same s and nearly the same state, and their errors are correlated. The p-values are therefore optimistic and descriptive.
  - They are also conditional on one training set per split; another position of the hold-out can change the margin [4].
- Sensitivity (Table 3): the test RMSE of every configuration on every split, averaged over the 4 initialisation sets, and the choice of both rules when the cross-validation uses other partitions. Neither is used for the selection.
- Reproducibility: every fit runs with one BLAS thread.
  - With the same library versions (scikit-learn 1.7.2, NumPy 2.2.5, OpenBLAS 0.3.30), a refit reproduces every number bit for bit. The sensitivity check refits the chosen configurations in other processes and matches every per-run RMSE exactly.
  - With several threads, L-BFGS stops at a slightly different point. Compared with an earlier run on 8 threads, a split's mean moved by up to 0.0003 K, one initialisation set's mean by up to 0.0013 K, and a single run by up to 0.008 K. Differences of that size are not meaningful.
- Cost: a fit takes seconds on one core, and a prediction microseconds. On 8 cores the evaluation takes 46 minutes, mostly the repeated cross-validation and the 100 leave-one-out refits; the sensitivity check takes 28 minutes.

split	held out / train	network	range	median	GP on s	GP on z	RBF	nearest	mean state
scattered	10 / 90	**0.017**	0.0008	0.013	0.019	0.022	0.021	0.038	0.161
band (three rows)	30 / 70	**0.038**	0.0010	0.033	0.045	0.051	0.057	0.118	0.204
centre 5×5	25 / 75	**0.025**	0.0004	0.019	0.029	0.035	0.037	0.080	0.162
centre 7×7	49 / 51	**0.029**	0.0006	0.026	0.035	0.069	0.068	0.116	0.203
leave-one-out	100 / 99	**0.024**	0.0004	0.017	0.028	0.032	0.034	0.085	0.221
top row	10 / 90	**0.049**	0.0028	0.033	0.079	0.083	0.084	0.176	0.482
outer 51	51 / 49	**0.067**	0.0005	0.027	0.074	0.081	0.093	0.134	0.265
outer 75	75 / 25	**0.114**	0.0009	0.041	0.125	0.131	0.138	0.159	0.226

Table 1: RMSE (K) of the last-year temperature over the water cells, mean over the held-out runs. Network: mean over the 4 initialisation sets; range: how much that mean varies across them; median: over the runs. Bold: lowest mean per split. The four-run split is in Section 6.

	one-standard-error rule (reported)			minimum CV RMSE				
split	difference [95 % interval]	p	Holm p	choice	network	difference	p	Holm p
scattered	-0.0025 [-0.0053, -0.0005]	0.037	0.074	10^{-3}	0.016	-0.0030	0.049	0.195
band (three rows)	-0.0075 [-0.0132, -0.0020]	0.015	0.044	10^{-3}	0.037	-0.0083	0.011	0.068
centre 5×5	-0.0045 [-0.0071, -0.0019]	0.003	0.013	10^{-4}	0.026	-0.0032	0.022	0.110
centre 7×7	-0.0058 [-0.0096, -0.0021]	0.003	0.013	10^{-3}	0.035	$+0.0010$	0.745	1.000
leave-one-out	-0.0046 [-0.0086, -0.0015]	< 0.001	< 0.001	10^{-3}	0.023	-0.0052	< 0.001	< 0.001
top row	-0.0297 [-0.0768, $+0.0001$]	0.193	0.193	10^{-4}	0.058	-0.0204	1.000	1.000
outer 51	-0.0076 [-0.0123, -0.0032]	0.002	0.013	10^{-3}	0.067	-0.0072	0.009	0.066
outer 75	-0.0111 [-0.0170, -0.0059]	< 0.001	0.004	10^{-2}	0.122	-0.0032	0.594	1.000

Table 2: Network minus GP on s, per held-out run (K; negative = network better): the mean difference, its 95 % bootstrap interval, the Wilcoxon signed-rank p and the Holm-corrected p over the eight comparisons. All are per-run and treat runs as independent, so they are optimistic (Section 5). Left: the network as reported (one-standard-error rule). Right: the network chosen by the minimum CV RMSE instead (ReLU, with the α given), and its mean RMSE.

	configurations within one standard error					choice, other partitions	
split	number	test RMSE	below GP on s	rank of the choice	all ten	one SE	minimum
scattered	5	0.016–0.017	5	3	0.016–0.034	same	same
band (three rows)	6	0.037–0.040	6	4	0.037–0.072	same	10^{-2}
centre 5×5	5	0.025–0.030	3	1	0.025–0.046	same	10^{-3}
centre 7×7	6	0.029–0.045	1	1	0.029–0.060	same	10^{-4}
top row	5	0.049–0.058	5	1	0.049–0.146	same	10^{-3}
outer 51	5	0.067–0.077	3	1	0.067–0.121	same	same
outer 75	6	0.114–0.144	3	1	0.114–0.165	same	same
four runs	6	0.117–0.187	2	5	0.098–0.187	–	–

Table 3: Sensitivity to the configuration. Left: the configurations whose CV RMSE is within one standard error of the best, the range of their test RMSE (mean over the 4 initialisation sets), how many beat the GP on s, and the rank of the rule's choice among them (1 = lowest test RMSE). All ten: the range over the whole grid. Right: the choice of each rule when the cross-validation repeats use other partitions ("same": unchanged; four runs: 5-fold is leave-one-out, so there is no other partition).

5 Findings

- **In interpolation the network has the lowest mean error of the methods tested** (Tables 1 and 2, Figure 2).
  - Against the GP on s the margin is 0.0025–0.0075 K, 13–17 % of the GP's error. It is consistent: lower on all five evaluations.

  - Per run, after the Holm correction, it is significant on band, both centre blocks and leave-one-out ($p \leq 0.044$), not on the scattered split ($p = 0.074$; 7 of 10 runs better). These tests are optimistic. With diagonals as units, none of the eight main evaluations is significant after the Holm correction†.
  - Linear interpolation of the fields along s, which fits nothing, appears to be as accurate as the network in interpolation and better than the GP on $s^{\dagger}$. So in interpolation most of the accuracy comes from the collapse onto s, not from the network.
  - Its mean error is 0.25–0.44 times the nearest run's. That ratio uses the first of the equidistant neighbours and is larger when they are averaged†.
- **In extrapolation its mean error is lower on all three splits**.
  - Per run the margin is significant on outer 51 and outer 75 (Holm $p = 0.013$ and 0.004), not with diagonals as units†. On outer 75 it holds only with the one-standard-error choice ($\alpha = 0.1$).
  - On outer 75 nearly all of the margin comes from held-out runs above the warm end of the training range in $s^{\dagger}$. The response steepens there; the GP flattens beyond its data, while the network continues the local slope, though still too shallow. A response that bends the other way could reverse this.
  - On outer 51 and the top row the margin comes mostly from $s_{\perp}$, which the GP on s cannot use†.
  - On the top row the gain (0.049 vs 0.079 K) is mostly the warm-corner run (0.16 vs 0.37 K), and the test is not significant ($p = 0.19$). Without that run, the network gets 0.037 vs 0.046 K and wins 6 of 9.
- **The margin over the GP on s depends on the selection rule** (Table 2, right).
  - The minimum CV RMSE picks a weaker regularisation. That network is still lower than the GP on s on 7 of the 8 evaluations.
  - It ties on the 7×7 block (+0.0010 K, $p = 0.75$) and on outer 75 (−0.0032 K, $p = 0.59$).
  - Per run, after the Holm correction, only leave-one-out remains significant; band and outer 51 come closest ($p = 0.068$ and 0.066).
  - Its error stays below the GP on z, the RBF and the nearest run on every main evaluation.
  - With other cross-validation partitions the one-standard-error choice is unchanged on all eight splits, and the minimum changes on four (Table 3).
  - The rule was adopted after the first review had seen test errors of other configurations (Section 3), so the margins under it are not a blind test.
- **Initialisation matters little, the configuration more** (Tables 1 and 3).
  - Four initialisation sets move a split's mean by 0.0004–0.0010 K, and by 0.0028 K on the top row.
  - Five or six configurations per split are within one standard error of the best CV score. Their test errors span up to 0.03 K (outer 75: 0.114–0.144 K).
  - On scattered, band and the top row, all of them beat the GP on s. On the 5×5 block, outer 51 and outer 75, 3 of them do; on the 7×7 block, 1 of 6.
  - The rule's choice has the lowest test error of the set on five of the seven splits, on the 5×5 block and outer 51 by less than the initialisation range. It ranks third of 5 on scattered and fourth of 6 on band.
- **In interpolation the error sits in the upper** 500 m (Figures 3 and 4).
  - It is 0.014–0.055 K there and 0.003–0.016 K below 1000 m.
  - At the centre of the 7×7 block, the predicted anomaly (relative to the training mean) has correlation 0.98 with the true one and amplitude 1.05.
  - This and the other example values are those of Figure 4, which shows the first initialisation set.
- **The warm edge is the hardest region**, and the corner ($c_k = 0.8$, $c_\epsilon = 0.0875$) the hardest run (Figure 4).
  - One row from the training set (top row), the network reproduces the corner's warming: correlation 0.99, amplitude 0.99, 0.157 K.
  - Three cells away (outer 75), it keeps a weak version of the pattern (correlation 0.62) with a third of the amplitude (0.33): a cold bias of −0.68 K and 1.052 K RMSE, mostly below 400 m.
  - Averaged over the four initialisation sets, the network's error there is 1.06 K. Every method is between 1.06 and 1.45 K.

- **Part of the extrapolation error is the basis.** Projecting the held-out truth on the training principal directions already leaves 0.010–0.017 K in interpolation and 0.031–0.079 K in extrapolation. On outer 75 this floor is spread along the warm edge†; the corner run's own floor is 0.85 K. Every method that predicts in this span shares it [16].
- **Extrapolated fields can be statically unstable†.** Toward the warm corner the predicted columns contain temperature inversions of up to several kelvin, against at most 0.17 K in the truth; with this equation of state they are density inversions. All methods show them, and so does the projection of the truth on the training basis, so the cause is the decoder. In interpolation the inversions are about 0.1 K, in the weakly stratified upper 500 m.
- **The spread over initialisations is not an uncertainty.** It is 0.003–0.009 K in interpolation and 0.008–0.011 K in extrapolation, while the error grows from 0.017 to 0.114 K.
- The comparison is between a 5-member ensemble and single baselines, and only the network's hyper-parameters are tuned by cross-validation.

6 Four training runs: cross-validation cannot choose

- Training on $c_k \in \{0.0198, 0.504\} \times c_\epsilon \in \{0.139, 3.53\}$ leaves 96 runs to predict, with $K = 3$ principal directions (2 inside the cross-validation folds).
- Cross-validation cannot choose here.
  - With four runs the 5-fold split is leave-one-out, so the three repeats are identical.
  - The standard error (0.15 K) covers six configurations.
  - Both rules pick ReLU with $\alpha = 1$, which also has the minimum CV RMSE.
- With that choice the network gets 0.147 K (0.134–0.169 over the initialisation sets). The GP on s gets 0.120 K, the RBF 0.146 K, the nearest run 0.181 K.
- Over the ten configurations the network ranges from 0.10 to 0.19 K (Table 3). Among the six within one standard error, the choice ranks fifth. The outcome is set by a choice that four runs cannot inform.
- Regimes:
  - on the flat side ($s < -2$, 46 runs), the medians are 0.040 (nearest run), 0.058 (network) and 0.061 K (GP on s);
  - on the warm side (50 runs) they are 0.197 (network), 0.172 (GP on s) and 0.223 K (nearest run).
- With $\alpha = 1$ the network is too flat (Figure 4, rows 2 and 5; first initialisation set).
  - At the centre run its anomaly has the right pattern (correlation 0.98) with 45 % of the amplitude: 0.114 K (0.124 K over the four sets), against 0.036 K for the GP on s.
  - At the corner: correlation 0.86, amplitude 0.40, 0.801 K (0.831 K over the four sets).

7 Next steps

- Bring the review's checks (†) into `det/`: paired tests with diagonals as units; linear interpolation along s, the nearest run with tied neighbours averaged, and a GP on $(s, s_\perp)$ as baselines; the errors by regime and by distance in s; a static-stability check.
- Inputs $(s, s_\perp)$ instead of $(\log c_k, \log c_\epsilon)$, to build in the collapse that makes the GP on s strong with few runs.
- For extrapolation, either a decoder not restricted to the span of the training states, or more runs along the warm edge (large c_k, small c_ϵ). Predicted columns need a convective adjustment, or a decoder that keeps them stable, before any use as model states.
- An uncertainty from resampling the training runs rather than the initialisations.
- Tune the baselines by cross-validation as well (RBF smoothing, GP noise and mean function), for an even comparison.
- Fix the selection rule, or α itself, in advance, and test it on new runs or new splits. Here the rule was revised after test errors had been seen, and with very few runs cross-validation cannot choose at all.

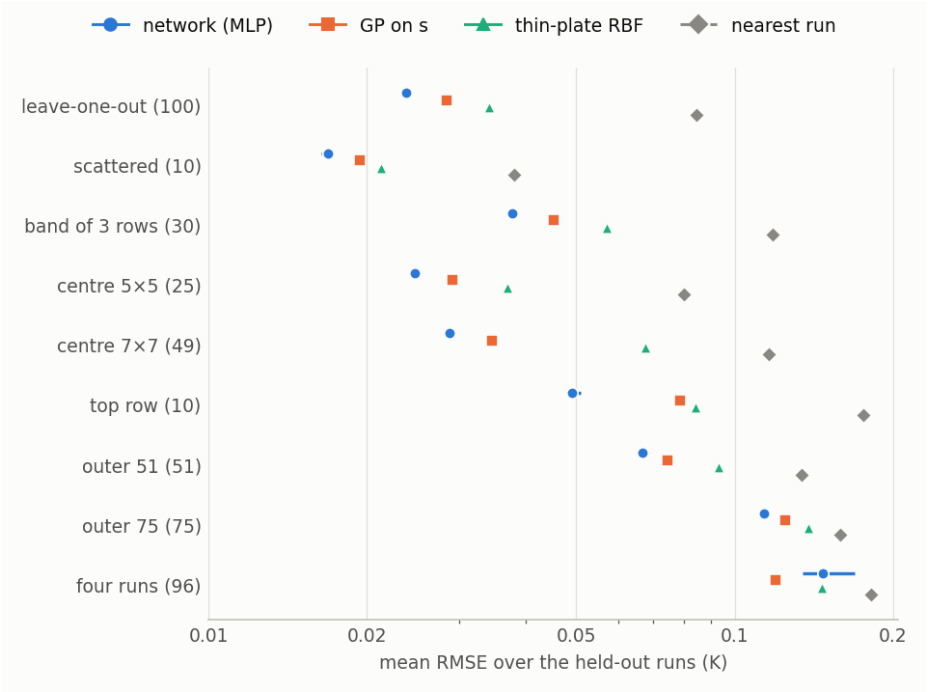


Figure 2: Mean RMSE over the held-out runs per split (log scale) for the network and three baselines: the GP on s, the thin-plate RBF and the nearest run. Held-out run counts are in brackets. Network: mean over 4 initialisation sets; the bar is their range.

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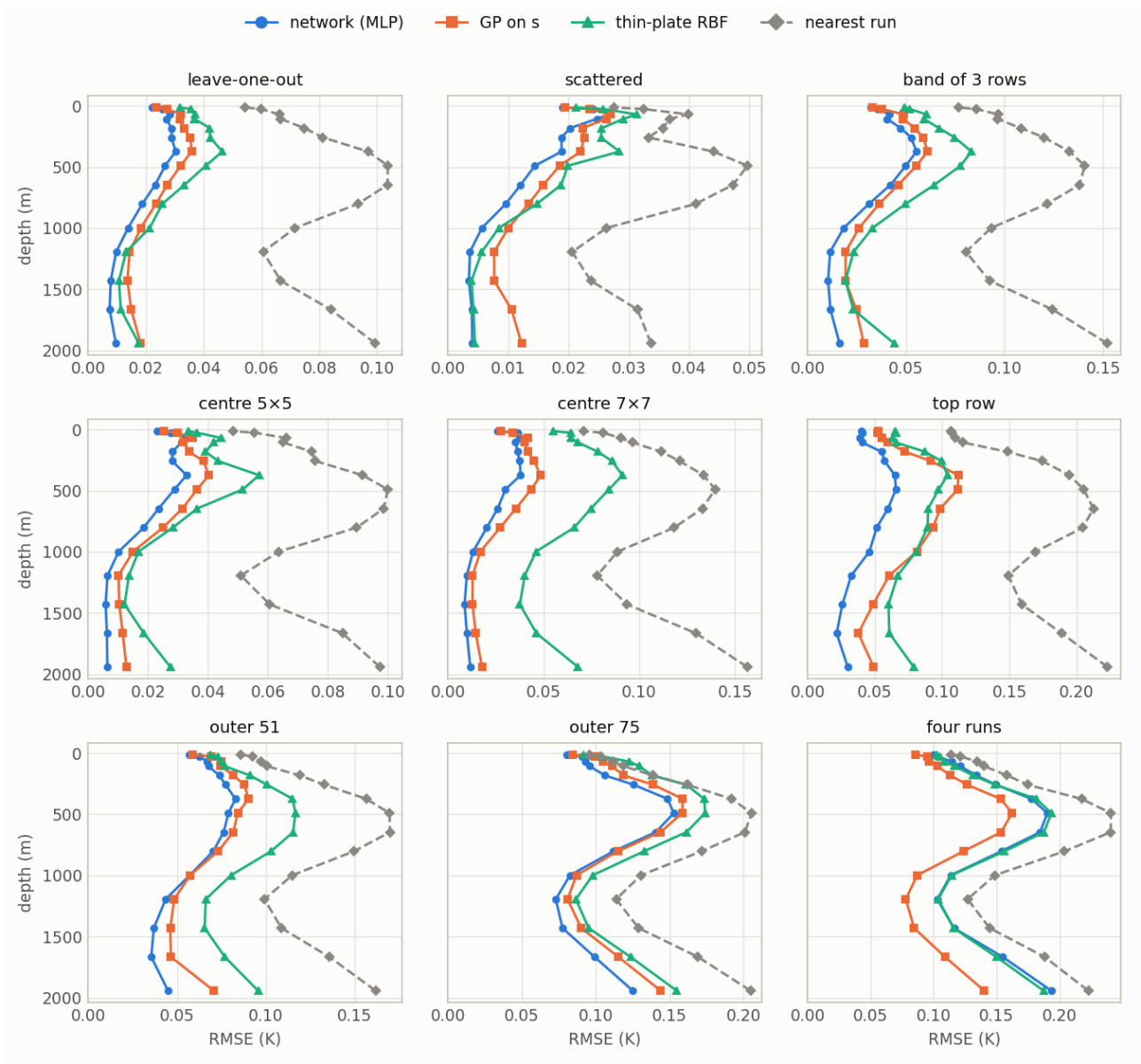


Figure 3: RMSE by depth, mean over the held-out runs of each split (network: averaged over the initialisation sets).

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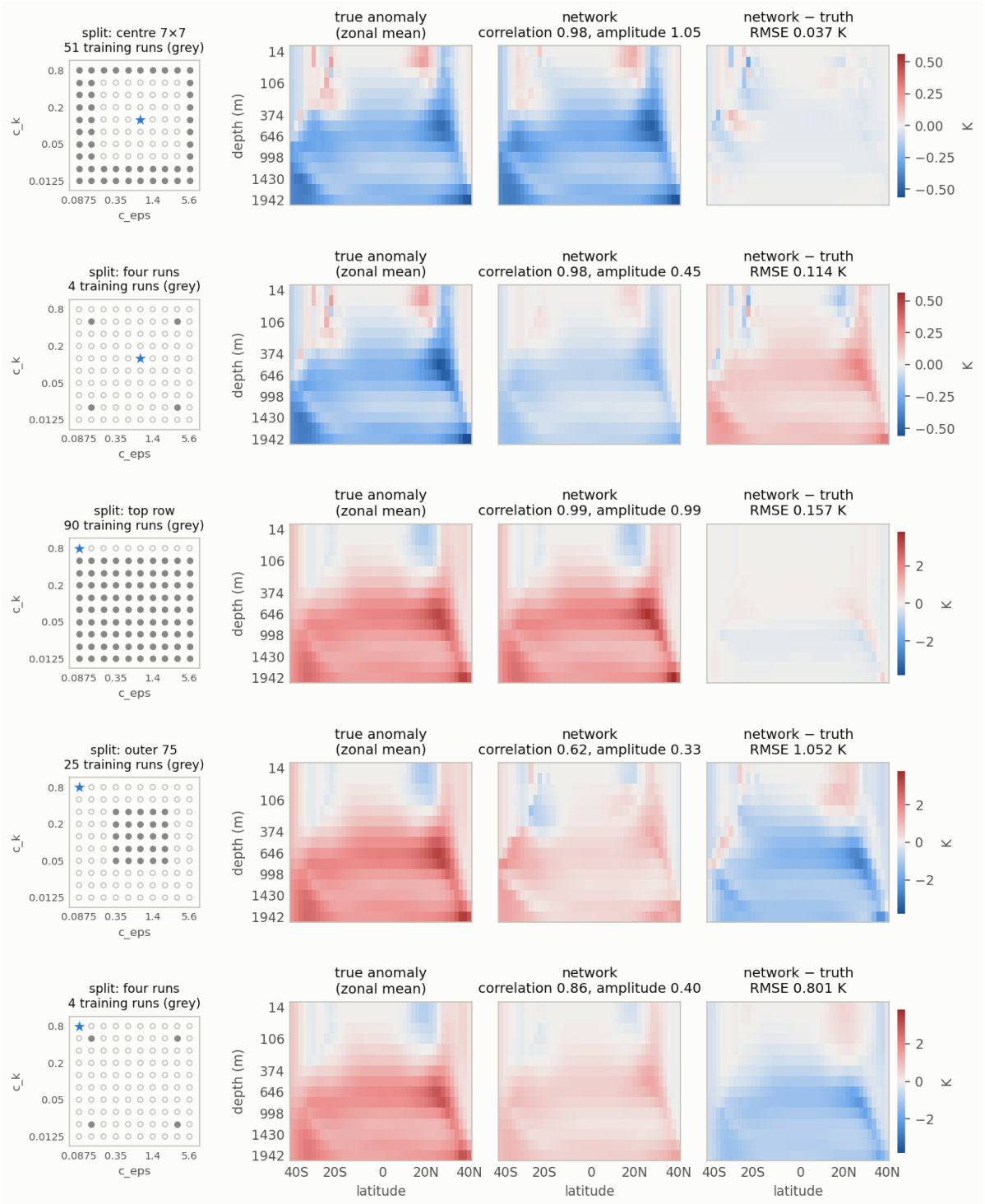


Figure 4: Zonal-mean sections of the last-year anomaly, relative to the mean of the split's training runs (what the network has to add to it): truth, network (first initialisation set) and their difference. Each run has one colour scale, and levels are drawn with equal height. Left: the split (grey: training runs; open circles: held out; star: the run shown). Rows 1–2: the centre run of the 7×7 block, held out with the block (51 training runs) and in the four-run split. Rows 3–5: the warm-corner run, held out with the top row (90 training runs), with the outer 75 (25) and in the four-run split (4). Correlation and amplitude (the regression slope of the predicted on the true anomaly) are over all water cells.